

WESTCLIFF UNIVERSITY · WEB603

TaskFlow

Personal Task Manager — Project 3

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Course: WEB603 — Full Stack Web Development

Module 5 · Project 3

PROJECT 3

Task 1

Project Proposal

Introduction · Goals · Target Audience · Key Features · Technology
Requirements

Introduction

TaskFlow is a full-stack Single Page Application (SPA) designed to help individuals organize and manage their daily tasks effectively. Users can create, update, prioritize, and complete tasks through a clean and responsive interface.

The app addresses a common problem: the lack of a simple, personal, and secure task management tool that works across devices without requiring a complex setup. TaskFlow delivers a modern web experience with user authentication, real-time data updates, and offline capability through Progressive Web App (PWA) technology.

The project is built as a full-stack application where **React** powers the front-end and **Express + MongoDB** powers the back-end, following industry-standard practices for security, scalability, and maintainability.

Goals & Objectives

PRIMARY GOALS

- Build a fully functional CRUD SPA for personal task management
- Provide secure signup, login, and logout with JWT authentication
- Enable users to add, edit, delete, and complete tasks
- Support task prioritization (High / Medium / Low)
- Deliver a responsive UI accessible on mobile and desktop

TECHNICAL OBJECTIVES

- Implement React with hooks and Context API
- Build a RESTful API with Express and MongoDB
- Secure routes with JWT tokens stored in HTTP-only cookies
- Apply SASS for modular, maintainable styling
- Enable PWA for installability and offline support
- Deploy on a live server with SSL/HTTPS

Target Audience

PRIMARY USERS

- **Age:** 18–45
- **Gender:** All genders
- **Occupation:** Students, freelancers, remote workers, professionals
- **Tech comfort:** Moderate to high — regular smartphone and web app users
- **Device:** Desktop, tablet, and mobile

USER NEEDS

- A private, personal space to track daily tasks
- Simple and fast interface — no learning curve
- Ability to prioritize tasks and filter by status
- Secure account so data is private to each user
- Works on any device, even with limited connectivity (PWA)

Key Features

- **User Authentication** — Secure signup, login, and logout with JWT stored in HTTP-only cookies and server-side sessions
- **Task CRUD** — Create, read, update, and delete personal tasks; each task belongs exclusively to the logged-in user
- **Priority Levels** — Assign High, Medium, or Low priority to each task with color-coded visual indicators
- **Task Status & Filtering** — Mark tasks as complete/incomplete; filter view by All, Active, or Done
- **Form Validation** — Client-side and server-side validation on all forms with clear error messages
- **Responsive Design** — Bootstrap 5 grid adapts layout for mobile, tablet, and desktop viewports
- **PWA Support** — Web App Manifest + Service Worker enable installation on device home screen and offline browsing
- **Live Stats** — Real-time counter showing total, active, and completed task counts

TASK 1 — PROPOSAL

Technology Requirements

Front-End

React 18

React Router v6

Axios

Bootstrap 5

SASS / SCSS

PWA (Manifest + SW)

Back-End

Node.js

Express.js

MongoDB

Mongoose

bcryptjs

JSON Web Token (JWT)

express-session

cookie-parser

Infrastructure & Tools

SSL / HTTPS

GitHub

Live Server (Render / Railway)

PROJECT 3

Task 2

Wireframes

Screen 1: Home (Task Dashboard) · Screen 2: Login · Screen 3: Sign Up

Home · Task Dashboard (Protected Route)

TaskFlow

Hello, [User Name]Logout

+ New Task

TITLE *

DESCRIPTION

PRIORITY

Total: 5 Done: 2 Active: 3

My Tasks

AllActiveDone

☐

High☐☐

☐

Med☐☐

☐

Low☐☐

Screen elements: Navigation bar (logo + user greeting + logout) · Stats bar (total/done/active counts) · Filter buttons (All / Active / Done) · Task form (title, description, priority selector, submit) · Task cards (checkbox, title, description, priority badge, edit + delete buttons) · Color-coded left border by priority

Login Page (Public Route)

The wireframe shows a login page for 'TaskFlow'. The browser's address bar displays 'https://taskflow.app/login'. The header is dark blue and contains the 'TaskFlow' logo on the left and 'Login' and 'Sign Up' buttons on the right. The main content area is light gray and features a centered white card with a dark blue border. The card is titled 'Welcome Back' and contains two input fields: 'EMAIL' and 'PASSWORD'. Below the password field is a dark blue submit button. The card also has a dark blue footer bar.

Screen elements: Navigation bar (logo, Login + Sign Up links) · Centered auth card with top accent border · Email input · Password input · Submit button · Link to Signup page · Inline validation errors shown below each field on submit

Sign Up Page (Public Route)

The wireframe shows a web browser window with the URL `https://taskflow.app/signup`. The header is dark blue and contains the TaskFlow logo on the left and 'Login' and 'Sign Up' buttons on the right. The main content area is light blue and features a centered white card titled 'Create Account'. The card contains three input fields: 'FULL NAME', 'EMAIL', and 'PASSWORD (MIN 6 CHARS)'. Each input field is a light gray rectangle with a thin border.

Screen elements: Navigation bar · Centered auth card · Full Name input · Email input · Password input (min 6 chars) · Confirm Password input · Submit button · Link to Login page · Client-side validation: required fields, email format, password match, min length